Analytical Chemistry Cumulative Examination February 15, 2019

1. Diagram a Michelson interferometer and label all of its components. Explain the operation of such a device, with the aid of suitable, clearly labeled diagrams. What is the ZPD point and why is the signal observed at this point so much different than that at other points? (25 points)

2. Please compare Raman and infrared spectroscopy with respect to: a) detectors used; (b) optics used; (c) solvent requirements; (d) wavelength range; (e) spectral selection rules; relationship between sample concentration and spectral peak intensity (55 points)

3. Two common detectors for infrared spectroscopy are DTGS and MCT detectors. What do the two names stand for? What is the common use(s) for each? Describe the operating conditions for each detector. (10 points)

4. Fourier transform infrared spectrometry has had a major impact on the practice of infrared spectrometry. Why is this so? What advantages does the technique provide when compared with traditional scanning IR? Please be specific. (10 points)